

YEAR 7 MATHEMATICS

Chapter 1

Integers

LEARN IT · PRACTISE IT

1

How to use this booklet

1 **Page 1: learn it.** Start with the **Do now**, from memory. Then read each worked **Example** and do the **Your turn** beside it. Each Your turn gives you a little less help.

2 **Check.** Everyone answers the one question at the end of page 1. It tells your teacher who is ready for page 2.

PAGE 2 **Spot the pattern, then Mixed practice:** today's skill mixed with earlier ones, so you have to choose the method. Finished? **Go further.**

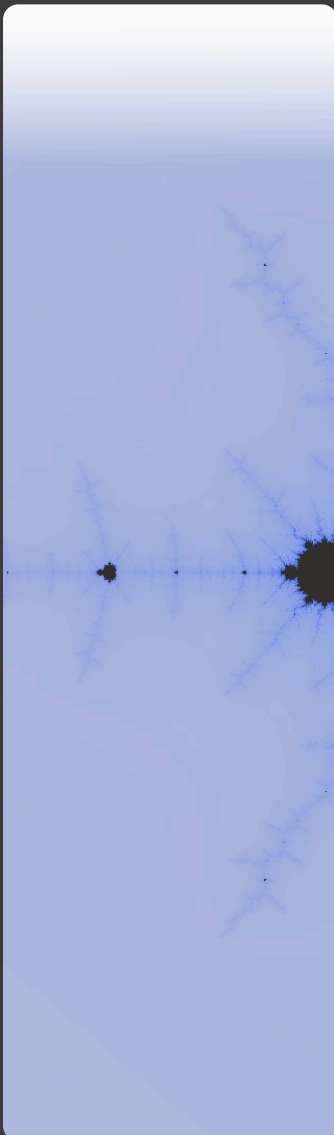
EXIT The **Exit ticket** is on your own. Circle how you went: Not yet, Nearly or Got it.

Contents and progress tracker

Lesson	Topic	Page	Date	Done
1.04	Adding integers	3		☐

In this chapter you will

- understand positive and negative numbers and their opposites
- use a number line to show and compare integers
- order integers
- add, subtract, multiply and divide integers
- use the order of operations with integers, with and without a calculator
- solve problems involving integers



About the artwork

The pictures in this booklet are parts of the **Mandelbrot set**, drawn by a computer that repeats one simple rule, $z \rightarrow z^2 + c$, thousands of times for every point. Every booklet shows a different part of the set: this one is the long antenna of the set along the negative real axis, dotted with tiny copies of the whole set. Maths can make beautiful things.

Acknowledgement of Country

We acknowledge the Bundjalung people, the Traditional Custodians of the land on which Kingscliff High School stands, and pay our respects to Elders past and present.

1.04 Adding integers

YEAR 7 · CHAPTER 1: INTEGERS

NAME _____ CLASS _____ DATE _____

DO NOW From memory. No notes.

1 $-7 \square -2$ Write $<$ or $>$.

1.03 2 Order smallest to largest: 4, -6, 0, -1

1.01 3 What is the opposite of -9?

1.02 4 Start at -4. Move 6 right. Where do you land?

THINK FIRST Have a go before we learn it. It's fine to be unsure.

It is -3°C at 6 am. By noon it is 5°C warmer. What is the temperature at noon?

WORDS

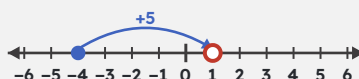
sum: the answer to an addition.

-3: say "negative three".

zero pair: +1 and -1 make 0.

BIG IDEA

Start at the first number. Adding a positive: move right. Adding a negative: move left.

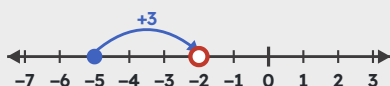


LEARN IT

Read each example. Then do the one beside it. Stuck? Look back at the example.

EXAMPLE 1

$$-5 + 3$$



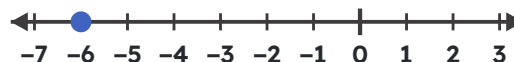
start at -5, move 3 right

$$= -2$$

Why? We moved right. Why?

YOUR TURN 1

$$-6 + 4$$

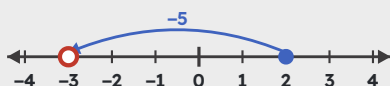


start at ____, move ____ right

$$= ______$$

EXAMPLE 2

$$2 + (-5)$$



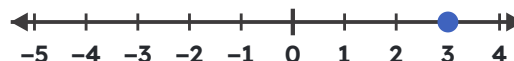
start at 2, move 5 left

$$= -3$$

Why? $2 - 5$ also moves 5 left from 2. What does that tell you about $2 + (-5)$?

YOUR TURN 2

$$3 + (-7)$$



start at ____, move ____ left

YOUR TURN 3

$$-2 + (-6)$$

EXAMPLE 3

$$-4 + (-3)$$

$$= -4 - 3$$

$$= -7$$

Why? Tom says $-4 + (-3) = 7$. What went wrong?

CHECK Everyone answers. Circle one letter, then hold it up.

Work out $-2 + 5$

(A) -7 (B) 3 (C) -3 (D) 7

SPOT THE PATTERN

Each question changes one thing. Work them out, then look down the answers.

1 $-3 + 3 =$ _____

2 $-3 + 2 =$ _____

3 $-3 + 1 =$ _____

4 $-3 + 0 =$ _____

5 $-3 + (-1) =$ _____

6 $-3 + (-2) =$ _____

7 $-3 + (-3) =$ _____

8 $-30 + (-30) =$ _____

What do you notice? What happens to the answer each time the number added goes down by 1?

MIXED PRACTICE

Today's skill mixed with earlier ones. Read each question carefully: they are not all the same kind.

1 $-8 + 5$ _____

1.04

2 Which is larger: -8 or 5 ? _____

1.03

3 $6 + (-9)$ _____

1.04

4 How far apart are -8 and 5 ? _____

1.02

5 $-7 + (-2)$ _____

1.04

6 What is the opposite of -12 ? _____

1.01

7 $-15 + 15$ _____

1.04

8 $-6 + \square = -1$ _____

1.04

9 A diver is at -12 m. She rises 5 m. Where is she now? _____

1.04

10 Order smallest to largest: $-3 + 4$, -3 , $-3 + (-4)$ _____

1.03

GO FURTHER

Finished? Try these. Explain your thinking.

1 Always, sometimes or never true? "Adding a negative number makes the answer smaller." Explain.

2 Find two integers that add to -3 . Then find two more, where one of them is positive.

3 Start at -4 . Add 2 , 6 and -5 in any order. Where do you finish? Does the order matter?

EXIT TICKET

On your own. Hand it in.

1 $-7 + 3$ _____

2 $4 + (-9)$ _____

3 It is -6°C . It warms up 10°C . What is the temperature now? _____

Today I can add positive and negative integers.

Not yet

Nearly

Got it